

Short Questions

Write Occurrence of Actinides in Earth Crust?

The only two actinides found in appreciable quantities in earth's crust are thorium and Uranium.

Small quantities of plutonium and neptunium are present in Uranium ores. Actinium and protactinium occur as decay products of certain thorium and Uranium isotopes.

What is role of Actinides in daily life.

They (Actinides) are used in smoke detectors and gas mantles.

They are mostly used in nuclear weapons.

They are used as fuel in nuclear reactions.

Differences between Lanthanides and Actinides

Binding Energy of 4f are higher
Binding Energy of 5f are lower

4f electrons have greater Shielding effect
5f electrons have poor Shielding effect

Maximum oxidation state is +4
Maximum Oxidation State is +7.

Additional electron enters in 4f orbitals.
Additional electrons enter in 5f orbitals

They don't form complexes easily
they have greater ability to form complexes

Except promethium are non-radioactive
they are radioactive.

They don't form oxocation
they form oxo-cations like UO^{2+}

Magnetic properties are easily explained
not easily explain

they are mostly colorless in nature
they are mostly colored ions.

CFT says that

splitting of d-orbitals is

due to electrostatic interaction. origin of splitting of crystal field exclusively is more complex.

→ it explains stability of complex more Δ then stability is more.

→ it explains color & spectra of complexes.

→ it explains magnetic properties of complexes.

MOT

MOT shows that

origin of splitting

is more complex.

→ it predicts distribution of electrons in a molecule.

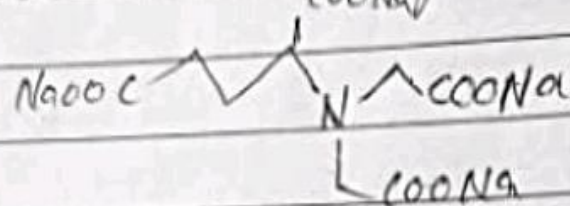
→ it helps predict molecular properties

such as shape,

magnetism and bond order.

5 What is role of chelates in Industry?

- i) In Industry chelates are use in detergent industry



- ii) Chelates are use in food industry for flavour retention, retard rancidity, prevents oxidation and discoloration.

6 Write name factors affect stability of M-complexes.

- i) Size of central metal ion.
- ii) Charge on metal ion.
- iii) Steric effect.
- iv) Chelation effect
- v) Rings Numbers
- vi) Hybridization.
- vii) Dipole moment. Effect.
- viii) Resonance Effect

3 What is Stability Constant?

The constant which describes the stability of a complex and also tells us those conditions and factors affecting it, is called stability constant.

- Generally, it describes a particular complex can be converted into other easily or not.
- Stability change with changing reagents and conditions.

What is Significance of Spectrochemical Series?

Spectrochemical series tells about splitting power of ligands.

It differentiate between strong and weak ligands.

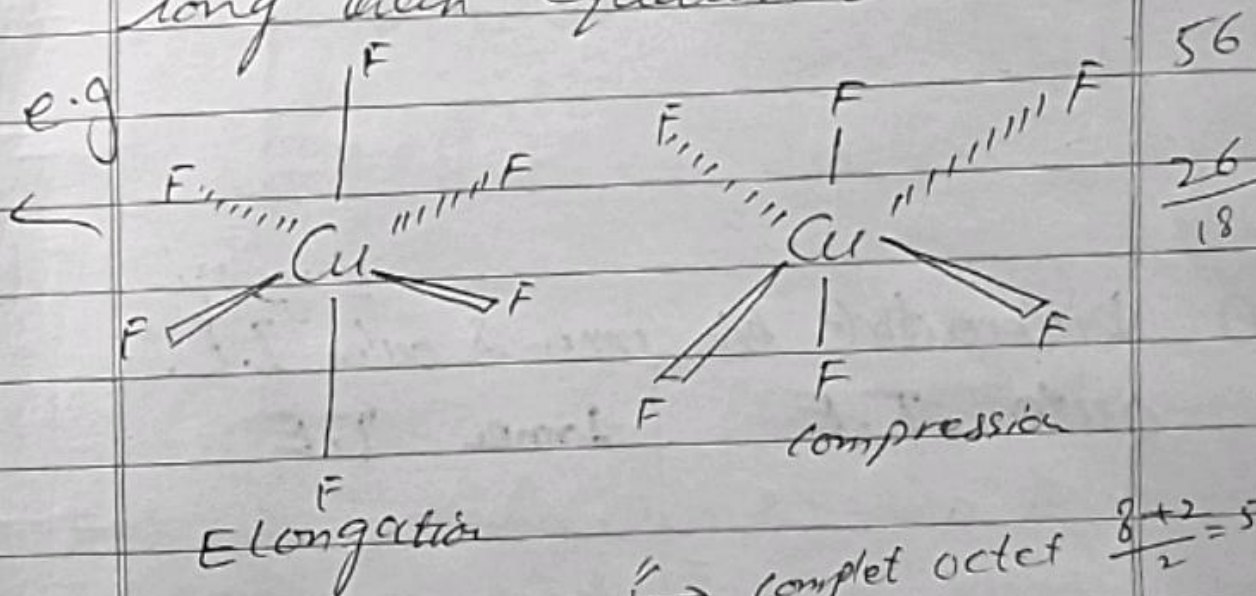
It differentiate between high and low spin complexes.

It tells that weak field ligands are π -donor and strong field ligands are π -acceptor ligands.

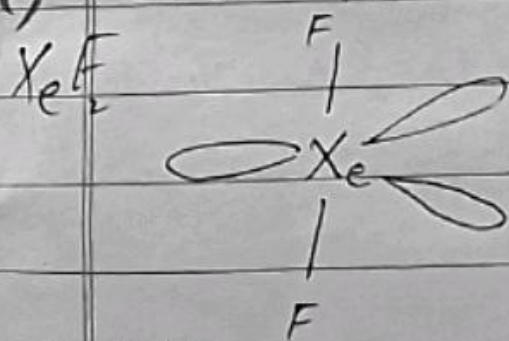
2020 Shout Qs

i) Explain John Teller distortion with example.

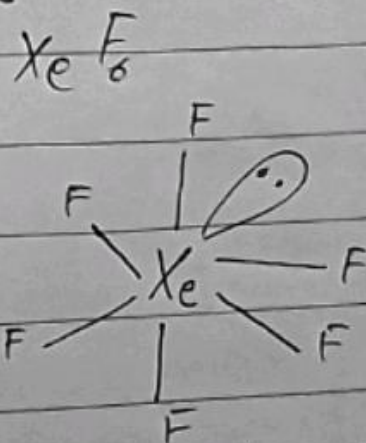
→ it is geometrical distortion of a non-linear molecular system that reduce its symmetry and energy. → it is ~~ense~~ observed b/c octahedral complexes where two axial bond can be shorter or long than equatorial bonds



2 time ii) What is shape of XeF_2 & XeF_6 molec.



shape linear



Distorted octahedral

iv) Differentiate b/w inner & outer T.E.

Outer T.E

d-block elements

are called outer T.E

e.g. Ni, Co, Fe, Cu
Zn

Group 3 $\rightarrow$ 12

Inner T.E

The elements in which last electron enters in f orbital.

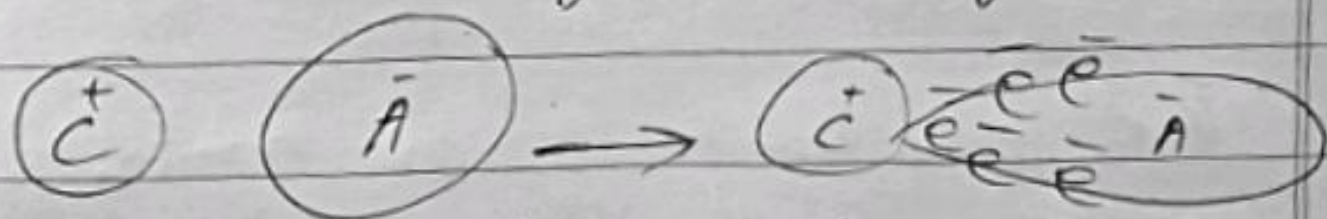
atomic no. 58-71, 90-103

e.g. La, Ac, Th, Pa, U
Md

1) What is Fajjan Rule.

→ This rule explain covalent character in ionic bond:

→ The reason is that ionic compound are not purely ionic due to polarization of anion and polarizing power of cation.



Cation withdraw electrons towards itself and electronic cloud of anion distorted from symmetrical shape.



SHOT ON itel
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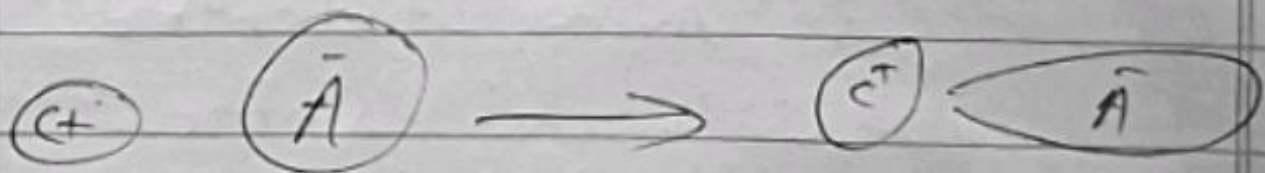
Differentiate b/w Polarizing Power and Polarizability?

Polarizing Power

The ability of a cation⁺ to polarize a nearby anion⁻ is called polarizing power.

Polarizability

The tendency of an anion to get distorted/polarized by cation is called polarizability.



ix) what is stability constant?

→ The constant which describes the stability of a complex and also tells us those condition and factors affecting it is called Stability constant.

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→ Stability change with changing reagents & conditions

x) How CFT is Superior to VBT?

→ CFT explain how orbitals split

→ CFT explain magnetic behaviour

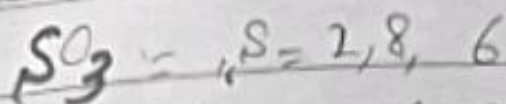
→ CFT explain outer & inner transition of complexes

* But VBT not explain these
So, CFT is superior

2019

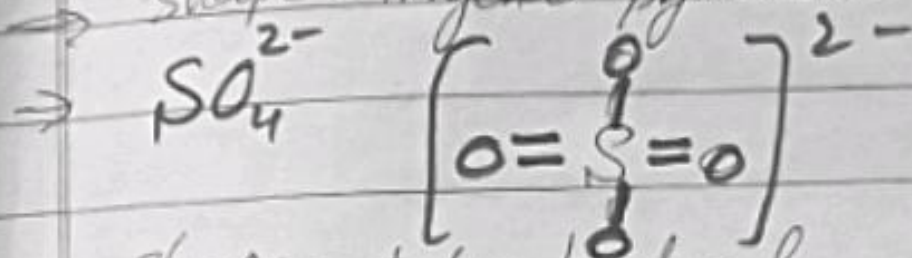
Short Qs

How VSEPR explain shape of SO_3 & SO_4^{2-}

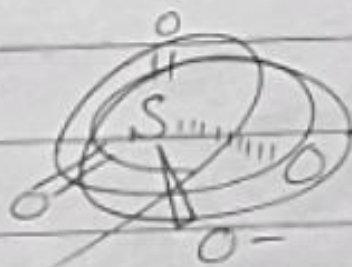


geometry = $\frac{6+3}{2} =$

Shape = Trigonal pyramidal



Shape = tetrahedral.



ii) Explain shape of AB_9 molecules with example

AB_9

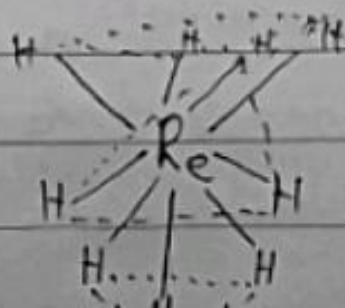
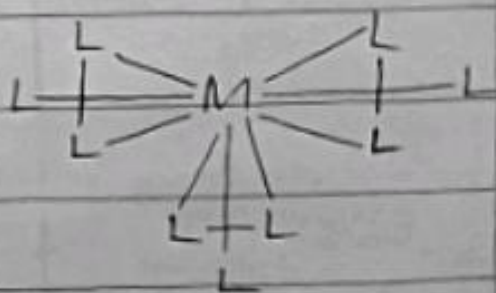
A = central atom

B = attached atoms

Geometry = Tricapped trigonal prism.

e.g ① ReH_9^{2-}

② TeH_9



(v) Repeat

(vi) Different b/e

Hybrid ~~orbitals~~ orbitals

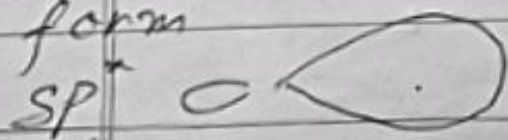
Molecular orbitals

when atomic orbitals
mix then new

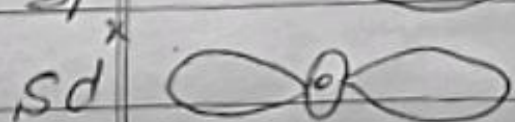
the orbital of two
different atoms

hybrid orbitals
form

merge to form new
orbital are called



molecular
orbital



greater overlap

low overlap

Stronger bond

weaker bond

more directional

less directional

what Applications of Uranium?

(U) uses

Date: _____

- it is used to generate ^{235}U .
- " " " in nuclear Stations
- " " " " color glass
- _____ to treat diseases
- _____ to kill cancer cells.

2017 Shout

- ① write down uses of Lanthanides.
- Use As misch metals
 - Use as catalysts.
 - " in color TV industry
 - " as refractories (耐火)

(ix) Define term Ferromagnetism?

→ it is a basic method in which a compound forms a permanent magnetic field.

eg Fe, Co, Ni, Steel.

2018 Short

- ① Why hybrid orbital exhibit better than simple atomic orbitals.

Reasons

- Hybrid orbitals are more directional.
- They have equal energy.
- They leads to greater overlap when bond form.
- They formed stronger bond

So



Q) what are post transition elements +
Give examples.

Date → The elements are present
b/w transition metals and
chemically weak non
metallic metalloids are called
→ They are weak π -back elements
→ P-block elements

(eg) Ga, In, Tl, Sn, Pb, Bismuth

Q. Can Lanthanide ion exist in +4 oxidation state? if yes then how, if no then why?

Ans. → NO,

→ This can justify on the basis of electronic configuration of Lanthanum.

→ La can lose only 3 e^- , 2s subshell & 1 e^- from d to attain noble gas e^- -configuration.

So _____

Inorganic
chem

Solved
by محمد زید